

LATU KALITA

Junior Research Fellow · School of Physics · IISER Thiruvananthapuram, Kerala 695551, India
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RESEARCH INTERESTS

Theoretical and computational condensed matter physics, centred on the interplay of **topology**, **non-Hermiticity** and **periodic driving**. My current work spans Floquet engineering of higher- and hybrid-order topological phases, non-Bloch band theory and the non-Hermitian skin effect, and the entanglement structure of driven, disordered and monitored fermionic systems. I am extending this towards measurement-induced transitions in Majorana circuits, open quantum systems and field-theoretic descriptions of topological criticality.

PUBLICATIONS AND PREPRINTS

Submitted / under review

[1] K. Roy, **L. Kalita**, B. Tanatar and S. Basu, *Floquet generation of hybrid-order topology and $\mathbb{Z}_2$ -like bipolar localization*, arXiv:2603.21954 (2026). Accepted modulo correction in *Physical Review Research*.

In preparation

[2] **L. Kalita**, K. Roy and S. Basu, *Floquet-driven superconducting diode effect in a helical Shiba chain* (manuscript in preparation, 2026).

RESEARCH POSITIONS

Junior Research Fellow, School of Physics, IISER Thiruvananthapuram Oct. 2025 – present
ANRF-funded project *Theoretical study of robust topological quantum systems and their characterization* · PI: Dr. Suraj S. Hegde

EDUCATION

M.Sc. in Physics · Indian Institute of Technology Guwahati 2023 – 2025
CGPA 8.99/10.0 · Thesis: *Non-Hermitian Topology and Floquet Skin Effect* · Advisor: Prof. Saurabh Basu

B.Sc. in Physics · Cotton University, Guwahati 2020 – 2023
CGPA 9.43/10.0 · First rank in the university · Gold Medal

RESEARCH EXPERIENCE

IISER Thiruvananthapuram · Supervisor: Dr. Suraj S. Hegde Oct. 2025 – present

- **Entanglement structure of driven and disordered Kitaev chains.** Numerical study of half-chain and topological entanglement entropy across disorder-induced transitions between Floquet Majorana phases; extraction of effective central charges and tests of infinite-randomness scaling.
- **Non-Hermitian topology in monitored and feedback-controlled systems.** Skin-effect-driven emergent Fermi surfaces in non-Hermitian fermionic chains, and measurement-induced entanglement transitions in Majorana circuits.

Indian Institute of Technology Guwahati · Supervisor: Prof. Saurabh Basu July 2024 – July 2025

- **Floquet hybrid-order topology and bipolar skin localization** [1]. Showed that periodic driving activates first-order helical edge states coexisting with higher-order corner modes in a Benalcazar–Bernevig–Hughes lattice, and identified a drive-controlled unipolar-to-bipolar skin localization transition. Characterized via a non-Bloch Floquet generalized-Brillouin-zone framework, real-space chiral winding numbers and spectral-localizer diagnostics.
- **Floquet-driven superconducting diode effect in a helical Shiba chain** [2]. Demonstrated a travelling-wave-drive-induced superconducting diode effect with tunable non-reciprocal critical currents and coexisting Floquet Majorana modes, requiring *no* external magnetic field; analysed within a self-consistent Bogoliubov–de Gennes formalism.
- **M.Sc. thesis.** Skin effect and the breakdown of conventional bulk–boundary correspondence in the Su–Schrieffer–Heeger model and in two-dimensional *p*-wave superconductors, using generalized-Brillouin-zone construction and Floquet theory.

Cotton University, Guwahati · Supervisor: Dr. Mridula Baro 2022 – 2023

- Synthesis of biomass-derived porous carbon nanomaterials by hydrothermal carbonization of bamboo powder with ZnCl_2 activation, for energy-storage applications.

TALKS, SCHOOLS AND WORKSHOPS

- 66th DAE Symposium on Nuclear Physics, 2022 (participant).
- NPTEL certification, *Numerical Methods and Simulation Techniques for Scientists and Engineers*, IIT Guwahati, 2021.

TECHNICAL SKILLS

- **Numerical methods.** Exact diagonalization; self-consistent Bogoliubov–de Gennes solvers; Floquet and Magnus expansions, two-frame stroboscopic evolution; non-Bloch band theory and generalized-Brillouin-zone construction; real-space topological markers and spectral localizers; entanglement entropy, inverse participation ratio and level-statistics computation; transfer-matrix methods.
- **Programming.** Python (NumPy, SciPy, Matplotlib), C/C++, Fortran, Mathematica, Scilab.
- **Tools.** $\text{\LaTeX}$

FELLOWSHIPS AND AWARDS

- **ANRF Junior Research Fellowship**, Anusandhan National Research Foundation, India — project *Theoretical study of robust topological quantum systems and their characterization* (2025 – present).
- **INSPIRE Scholarship**, Department of Science and Technology, India, for academic excellence in the basic sciences (2020 – 2025).
- **Gold Medal**, Cotton University — first rank, B.Sc. Physics (2023).

NATIONAL EXAMINATIONS

- **CSIR–UGC NET (Physics)**, December 2025 session — All India Rank 174 of 22,533. Qualified for the Junior Research Fellowship and Assistant Professor eligibility.
- **GATE (Physics)** 2025 — All India Rank 325.
- **IIT-JAM (Physics)** 2023 — All India Rank 271.

TEACHING

- **Private tutor**, Physics — one-to-one tuition, 2026 – present. Weekly sessions electromagnetism, Mathematical physics, Quantum Mechanics;

COURSEWORK AND LANGUAGES

Selected graduate coursework. Quantum Field Theory · General Relativity · Quantum Information and Computation · Open Quantum System · Topics in Condensed Matter Physics · Advanced Statistical Mechanics · Solid State Physics · Mathematical Physics

Languages. English (professional, medium of instruction throughout) · Hindi (native) · Assamese (mother tongue)

REFEREES

Dr. Suraj S. Hegde — current supervisor

Assistant Professor, School of Physics, IISER Thiruvananthapuram, Kerala, India · surajhegde@iisertvm.ac.in

Prof. Saurabh Basu — M.Sc. project and thesis supervisor

Professor, Department of Physics, IIT Guwahati, Assam, India · saurabh@iitg.ac.in